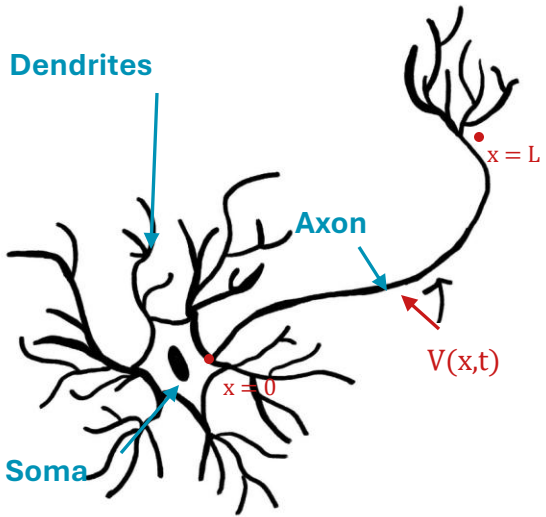
Distributed Hodgkin-Huxley dynamics

Single-neuron anatomy and a spatially distributed axon model



A local membrane state at every position x

$$C_M \partial_t V = \underbrace{\frac{r}{2R}\partial_{xx} V}_{\text{axial coupling}} + I_{\text{app}} - \underbrace{I_{\text{ion}}}_{\text{local currents}},$$

$$I_{\text{ion}} = g_K n^4 (V - V_K) + g_{\text{Na}} m^3 h (V - V_{\text{Na}}) + g_l (V - V_l),$$

$$\partial_t q = \frac{q_\infty(V) - q}{\tau_q(V)}, \qquad q \in \{n, m, h\}.$$

x: distance along the axon; n, m, h: local channel gates

Eq. (9): uniform unmyelinated axon. Dendrites and soma require a branched-cable extension.

Image: Vatika / Science for Everyone (2024). Model: Martinez et al. (2023), Eqs. (8)-(9), p. 708.